ADF Pipeline Documentation: GL Journal

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Table of Contents

1. Overview
2. Pipeline: GL_Journal
3. Pipeline: GL_Journal_pagination
4. Data Sources
5. Data Dictionary
6. Storage Layout
7. Linked Services & Datasets

Overview

These two pipelines together implement a paginated OData ingestion pattern for General Ledger Journal data into a medallion lakehouse architecture (Bronze → Silver → Gold). `GL_Journal` is the orchestrator pipeline that manages scheduling logic, archival of prior data, and downstream Databricks processing. `GL_Journal_pagination` is the child pipeline that handles chunked, paginated extraction from an OData source.

Architecture Flow:

```
OData Source (ERP)
      │
      ▼
GL_Journal_pagination  ← called per fiscal year/period
  (Paginated copy: 45,000 rows/page → Parquet files in Bronze Staging)
      │
      ▼
GL_Journal (Orchestrator)
  ├── Archive prior staging files → Bronze Archive
  ├── ForEach fiscal period → calls GL_Journal_pagination
  ├── Databricks: object_logs (Bronze)
  ├── Databricks: GL Journal Silver and Gold migration
  └── Databricks: object_logs (Silver/Gold)
```

Pipeline: GL_Journal

Type: Orchestrator
Trigger Logic: Frequency is determined by trigger name. If the trigger is `trg_gl_monthly`, frequency is `monthly`; otherwise it is `year` (quarterly, 3 periods).

Activity Flow

Step	Activity Name	Type	Depends On
1	`Set_calYear`	SetVariable	—
2	`Set_calMonth`	SetVariable	`Set_calYear`
3	`Set_currentFY`	SetVariable	`Set_calMonth`
4	`Set_previousFY`	SetVariable	`Set_currentFY`
5	`Set_currentFP`	SetVariable	`Set_previousFY`

Step	Activity Name	Type	Depends On
6	`Set_FYIndex`	SetVariable	`Set_currentFP`
7	`Set_StartMonth`	SetVariable	`Set_FYIndex`
8	`Set_fiscalYear_Array`	SetVariable	`Set_StartMonth`
9	`Get_Meta_staging`	GetMetadata	`Set_fiscalYear_Array`
10	`ForEach1`	ForEach (Sequential)	`Get_Meta_staging`
10a	`Archive_yesterdays_data_copy1`	Copy	— (inner)
10b	`Delete_staging_copy1`	Delete	`Archive_yesterdays_data_co`
11	`ForEachYearandPeriod`	ForEach (Sequential)	`ForEach1`

Step	Activity Name	Type	Depends On
11a	`loopFY`	SetVariable	— (inner)
11b	`loopFP`	SetVariable	`loopFY`
11c	`Execute Pipeline1`	ExecutePipeline	`loopFP`
12	`odata_bronze_log_NB`	DatabricksNotebook	`ForEachYearandPeriod` (Succe
12 (fail)	`odata_bronze_log_NB_fail`	DatabricksNotebook	`ForEachYearandPeriod` (Failed
13	`migration_to_silver`	DatabricksNotebook	`ForEachYearandPeriod` (Succe
14	`odata_silver_gold_log_NB`	DatabricksNotebook	`migration_to_silver`

Variables

Variable	Type	Default Value	Description
`source_file_name`	String	`factgljournal_`	Prefix used for staging Parque names.
`main_file_name`	String	`Main_data<date>_GL_Journals.parquet`	Name pattern the consolidat main file.
`source_file_path`	String	`finance/bronze/factGlJournals/staging`	ADLS path for Bronze staging
`archive_file_path`	String	`finance/bronze/factGlJournals/archive`	ADLS path for archived Bron files.
`main_file_path`	String	`finance/bronze/factGlJournals/main`	ADLS path for consolidated n files.
`archive_name_path`	String	`Raw_delta<date>_GL_Balances.parquet`	Archive filena pattern (note: references GL_Balances b default — may copy/paste art
`container_name`	String	`apm/`	Azure Data La Storage contai name.
`obj_name`	String	`factGlJournals`	Object name u logging noteb
`obj_id`	Integer	`23`	Object ID used logging noteb
`layer_bronze`	String	`bronze`	Layer label for bronze log ent
`layer_silver_gold`	String	`silver/gold`	Layer label for silver/gold log

Variable	Type	Default Value	Description
			entries.
`layer_gold`	String	`gold`	Gold layer labe (declared but r explicitly pass visible activiti
`odata_monthly_path`	String	`RPZDB952B08B8E04C42E5F69DQueryResults`	OData entity p for monthly frequency que
`odata_yearly_path`	String	`RPZ6E42D26A90645956B90252QueryResults`	OData entity p for yearly/quar frequency que
`frequency`	String	`monthly` or `year`	Determined by trigger name (`trg_gl_mor` → `monthly`, otherwise `yea`
`calYear`	Integer	`0`	Current UTC calendar year (computed at runtime).
`calMonth`	Integer	`0`	Current UTC calendar mont (computed at runtime).
`currentFY`	Integer	`0`	Current fiscal y (computed at runtime).
`previousFY`	Integer	`0`	Previous fiscal (computed at runtime).
`currentFP`	Integer	`0`	Current fiscal p 1–12 (compute runtime).

Variable	Type	Default Value	Description
`fiscalYearMonthIndex`	Integer	`0`	Linear FY-mo index = `(currentFY 12) + currentFP`.
`startMonth`	Integer	`0`	Start of the processing wi as a linear inde
`fiscalYears`	Array	`[]`	Array of linear month index integers to pro
`FiscalYear`	Integer	—	Loop variable: current iterati fiscal year.
`FiscalPeriod`	Integer	—	Loop variable: current iterati fiscal period.
`skip`	Integer	`0`	Pagination ski offset (mirrors pipeline state)
`counter`	Integer	`0`	Running coun rows copied (u log notebooks
`stop`	Boolean	`false`	Pagination sto (mirrors child pipeline state)

Pipeline: GL_Journal_pagination

Type: Child / Worker
Called By: `GL_Journal` → `ForEachYearandPeriod` → `Execute Pipeline1`

Activity Flow

Step	Activity Name	Type	Depends On	Description
1	`Reset Stop`	SetVariable	—	Initializes `stop = false` at the start of each invocation.
2	`Reset Skip`	SetVariable	`Reset Stop`	Initializes `skip = 0` to begin paginatior from the first record.
3	`Until1`	Until Loop	`Reset Skip`	Loops until `stop == true`. Each iteration fetches one page of up to 45,000 records.
3a	`odata to sink`	Copy	— (inner)	Queries the OData source with `$select` `$filter`, `$orderby`, `$skip`, and `$top=45,000`. Writes results as a Parquet file to ADLS Bronze staging. The filename encodes `FiscalYear_FiscalPeriod_<prefix <skip>.parquet`.
3b	`Set Stop`	SetVariable	`odata to sink`	Sets `stop = (rowsCopied < 45000)` If a full page was returned, more pages ma exist; otherwise, extraction is complete.
3c	`Increment_Skip`	SetVariable	`odata to sink`	Sets `counter = skip + rowsCopied` (intermediate accumulator).
3d	`Set Skip`	SetVariable	`Increment_Skip`	Promotes `counter` into `skip` for the next iteration. ADF requires a two-variable pattern for in-loop self-assignment.

Parameters

Parameter	Type	Description
`frequency`	String	`monthly` or `year`. Determines which OData path is used.
`FiscalYear`	Int	The fiscal year to query (e.g., `2025`).
`FiscalPeriod`	Int	The fiscal period (1–12) to query.

Parameter	Type	Description
`odata_monthly_path`	String	OData entity path for monthly data.
`odata_yearly_path`	String	OData entity path for yearly data.
`source_file_name`	String	Parquet filename prefix for staging output.
`source_file_path`	String	ADLS folder path for staging output.
`container_name`	String	ADLS container name.

Variables

Variable	Type	Default	Description
`skip`	Integer	`0`	Current OData `$skip` offset. Incremented each loop iteration.
`stop`	Boolean	`false`	Signals the Until loop to terminate when fewer than 45,000 rows are returned.
`counter`	Integer	—	Intermediate accumulator used to update `skip` (required due to ADF self-reference restriction).

OData Query Structure

```
$select=CFISCPER,CCOMPANY_UUID,CFISCYEAR,CGLACCT,CACC_DOC_UUID,CPOSTING_DATE,
        CPROFITCTR_UUID,CPROJ_TASK_UUID,KCBALANCE_CURRCOMP,CNOTE_HD,
        KCCREDIT_CURRCOMP,KCDEBIT_CURRCOMP
&$filter=CFISCYEAR eq {FiscalYear} and CFISCPER eq {FiscalPeriod}
&$orderby=CFISCYEAR desc, CFISCPER desc
&$skip={skip}
&$top=45000
```

Data Sources

Source	Type	Reference Name	Description
OData ERP (Monthly)	OData	`ODATA_PROD_Resources` with path `RPZDB952B08B8E04C42E5F69DQueryResults`	Production OData fe monthly entity path. OData service.

Source	Type	Reference Name	Description
OData ERP (Yearly/Quarterly)	OData	`ODATA_PROD_Resources` with path `RPZ6E42D26A90645956B90252QueryResults`	Production OData fe yearly/quarterly enti different OData entit
ADLS Bronze Staging	Azure Data Lake Storage Gen2 (Parquet)	`ParquetDS`	Staging zone where during ingestion (`finance/bronze/`
ADLS Bronze Archive	Azure Data Lake Storage Gen2 (Parquet)	`ParquetDS`	Archive zone for pric (`finance/bronze/` Files are date-stamp
ADLS Bronze Main	Azure Data Lake Storage Gen2 (Parquet)	— (path variable)	Consolidated main fi (`finance/bronze/` Referenced in variab in these pipelines (lil
Databricks (Silver/Gold Migration)	Azure Databricks	`AzureDatabricks2`	Notebook: `Finance` `files/GL Journal` from Bronze staging Delta tables.
Databricks (Object Logging)	Azure Databricks	`AzureDatabricks2`	Notebook: `Finance/Noteboo` Writes pipeline execu duration, errors) to a
GetMetadata (Staging Inventory)	Azure Data Lake Storage Gen2	`Parquet_Meta`	Read-only metadata enumerate existing f

Data Dictionary

OData Source Fields → Sink Column Mappings

Source Field	Source Type	Sink Column Name	Sink Type	Description
`CACC_DOC_UUID`	String	`DOC_ID`	String	Unique identifier for the accounting document (journal entry). UUID format.
`CCOMPANY_UUID`	String	`CCOMPANY`	String	Company identifier (UUID). Represents the legal entity or company code.
`CFISCPER`	Int64	`CFISCPER`	Int64	Fiscal period number (1–12). Used as a filter in the OData query.
`CFISCYEAR`	Int64	`CFISCYEAR`	Int64	Fiscal year (e.g., 2025). Used as a filter in the OData query.
`CGLACCT`	String	`CGLACCT`	String	General Ledger account code.
`CNOTE_HD`	String	`JE_HEADER_TEXT`	String	Journal entry header text / note. Descriptive label for the GL posting.
`CPOSTING_DATE`	DateTime	`CPOSTING_DATE`	DateTime	Date the journal entry was posted in the ERP system.
`CPROFITCTR_UUID`	String	`CPROFITCTR`	String	Profit center identifier (UUID). Used for cost center / profitability reporting.
`CPROJ_TASK_UUID`	String	`CPROJ_TASK`	String	Project and task identifier (UUID).

Source Field	Source Type	Sink Column Name	Sink Type	Description
				Links the journal entry to a project or work order.
`KCBALANCE_CURRCOMP`	Decimal	`KCBALANCE_CURRCOMP`	Decimal	Net balance amount in company currency.
`KCCREDIT_CURRCOMP`	Decimal	`KCCREDIT_CURRCOMP`	Decimal	Credit amount in company currency.
`KCDEBIT_CURRCOMP`	Decimal	`KCDEBIT_CURRCOMP`	Decimal	Debit amount in company currency.

Databricks Notebook Parameters — `object_logs`

Parameter	Description
`obj_id`	Integer ID of the pipeline object being logged (GL Journal = `23`).
`obj_name`	Object name string (`factGlJournals`).
`status`	Execution status: `Succeeded` or `Failed`.
`number_rows`	Row count processed (from `counter` variable or migration notebook output).
`duration`	Execution duration in seconds (from Databricks output or `0` for ADF-native activities).
`error_details`	Error message string, or `no error` if successful.
`layer`	Medallion layer label: `bronze`, `silver/gold`.
`storage_name`	Azure Storage Account name (from ADF global parameter `StorageAccountName`).

Databricks Notebook Parameters — `GL Journal Silver and Gold`

Parameter	Description
`IngestionFlag`	Boolean-as-string. `true` if it is the 8th of the month OR no prior staging files existed (triggers a full reload). `false` for incremental.

Parameter	Description
`storage_name`	Azure Storage Account name.
`catalog_name`	Unity Catalog name: `apm-nprd-adls-01`.
`fiscalYearMonthIndex`	Linear FY-month index passed to determine which periods to process in Silver/Gold.
`frequency`	`monthly` or `year`. Controls processing scope in the notebook.

Storage Layout

```
<container: apm/>
└── finance/
    └── bronze/
        └── factGlJournals/
            ├── staging/        ← Active ingest zone (paginated .parquet chunks)
            |   └── {FY}_{FP}_{prefix}{skip}.parquet
            ├── archive/        ← Prior-run files (date-stamped)
            |   └── {yyyy-MM-dd}_{FY}_{FP}_{prefix}{skip}.parquet
            └── main/           ← Consolidated main files (written by Databricks)
                └── Main_data{yyyy-MM-dd}_GL_Journals.parquet
```

Linked Services & Datasets

Name	Type	Usage
`ODATA_PROD_Resources`	OData Linked Service	Source dataset for ERP GL Journal OData feeds. The `odata_path` parameter selects the monthly or yearly entity at runtime.
`ParquetDS`	Azure Data Lake Storage Gen2 Parquet Dataset	Parameterized dataset used for both reading (staging/archive source) and writing (staging/archive sink). Accepts `file_name`, `file_path`, and `container` parameters.

Name	Type	Usage
`Parquet_Meta`	Azure Data Lake Storage Gen2 Parquet Dataset	Read-only metadata dataset pointing to the staging folder. Used by `GetMetadata` to enumerate existing files.
`AzureDatabricks2`	Azure Databricks Linked Service	Used by all three Databricks notebook activities: bronze logging, silver/gold migration, and silver/gold logging.

Notes & Observations

- **Fiscal Year Logic:** The fiscal year starts in July (month 7). Calendar months July–December belong to the *next* fiscal year (e.g., July 2024 = FY2025, Period 1).
- **Pagination Pattern:** The `GL_Journal_pagination` child pipeline uses ADF's two-variable counter pattern (`counter` → `skip`) as a workaround for ADF's restriction on using a variable to update itself within the same loop iteration.
- **Page Size:** Each OData page fetches up to 45,000 rows. The loop terminates when a page returns fewer than 45,000 rows.
- **Archive Naming:** The `archive_name_path` default value references `GL_Balances` rather than `GL_Journals`, which appears to be a copy/paste artifact from a related pipeline.
- **IngestionFlag Full Reload Trigger:** A full Silver/Gold reload is triggered on the 8th of each month or when the staging folder was empty at pipeline start.
- **Catalog Name:** The Unity Catalog name is hardcoded as `apm-nprd-adls-01` in the migration notebook parameters, suggesting a non-production environment catalog is targeted.